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Data-driven investment and performance management in the livestock sector

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Summary

Evidence-based decision-making is now axiomatic in many sectors and has become increasingly important in prioritising development in low- and middle-income countries. In the livestock development sector, there has been a lack of data on health and production required to establish an evidence base. Thus, much strategic and policy decision-making has been based on the more subjective grounds of opinion, expert or otherwise. However, there is now a trend towards a more data-driven approach for such decisions. The Centre for Supporting Evidence-Based Interventions in Livestock was established in Edinburgh by the Bill and Melinda Gates Foundation in 2016, to collate and publish livestock health and production data, lead a community of practice to harmonise livestock-data-related methodologies, and develop and monitor performance indicators for livestock investments.

Keywords

Data-driven investment – Decisions – Evidence-based decisions – Livestock – Livestock investment – Livestock performance management – Productivity.

Introduction

'With little or no knowledge regarding animal health, nutrition, production process, and reproduction procedures, and without a scientific plan and program, achieving a desired economic profit is absolutely impossible' [1].

Evidence-based decision-making has become broadly accepted as necessary for effective decision-making in many professional settings. This has perhaps been most notable in the field of human medicine, where it began to emerge as a new model for more rational clinical practice in the early 1990s [2], and has gained wide global acceptance since then [3, 4, 5, 6]. Evidence-based decision-making has since been adopted in veterinary medicine [7, 8, 9, 10] and many other professional and business sectors [11]. Evidence-based medicine has been defined as the 'integration of best research evidence with clinical expertise and patient values' [12] and this principle has been readily translated into other specific professional sectors.

The evolution of the livestock strategy in the Bill and Melinda Gates Foundation: a case study

A strategy for agricultural development in the smallholder sector of low- and middle-income countries (LMICs) was initiated by the Bill and Melinda Gates Foundation (BMGF) in 2006. It soon became apparent that most smallholders rely on both livestock and crops for their livelihoods. However, there was little understanding of their livestock development priorities due to a distinct lack of data-based information on the needs of this sector [13]. Therefore, a few initial investments were made to learn more and to help inform future livestock investments; this process then resulted in the launch of a livestock initiative in 2012.

Seven countries in sub-Saharan Africa and two in South Asia were chosen for the programme for dairy cattle (including water buffalo in South Asia), small ruminants and poultry, focusing on animal health and genetics, and the value chains of the respective species. Fourteen infectious diseases considered to be causing the highest productivity losses were selected for investment. In genetics, the focus was on identification and multiplication of appropriate genetic technologies for smallholders, particularly in dairy and poultry. In prioritising these investments, extensive consultation took place with partners and consultants to gain expert advice on the most effective direction to take. However, as learning and experience accrued over those early years, it became apparent that much of the input was based on experience and expert opinion, rather than objective evidence. This was largely because of the extensive gaps in field data on these livestock

health and productivity issues. This pragmatic rather than evidence-based approach to decision-making can be illustrated by the example of how livestock diseases were initially prioritised for intervention by BMGF. The 14 priority diseases referred to above were selected on expert opinion and largely based on the report by Perry *et al.* [13], commissioned by the United Kingdom's (UK) Department for International Development (DFID).

A not-for-profit organisation, the Global Alliance for Livestock Veterinary Medicines (GALVmed), originally the Global Alliance for Livestock Vaccines (GALV), was set up in 2005 with seed funding from the then UK DFID. This followed the realisation that there was a market failure in the provision of animal health products to small-scale producers, which applied particularly to vaccines for a large number of diseases. Thus, the need was identified for an organisation to support the development and production of vaccines, diagnostics and other products for the control of these neglected tropical livestock diseases. Whilst there had been a considerable body of research on these diseases over the years, the gap in the livestock vaccine market was perceived to be due to the unwillingness of the animal health industry to invest in those diseases because of the relatively disparate and supposedly low-demand and high-risk market segments. In 2008 BMGF then also invested in GALVmed along with DFID, this being the first major BMGF investment in animal health.

Early in the development of GALV/GALVmed, a committee of technical advisers comprising veterinary scientists experienced in international development and global vaccine developers, including one of the present authors, was convened to advise the organisation on disease prioritisation for its initial focus. The recommendation for 12 infectious diseases was subsequently adopted as a key priority for GALVmed. These decisions were largely based on the advice given by experts on the basis of their experience and knowledge of the sector, but with little or no direct input of solid data. Further diligence was carried out to support the decision, but it was noted at the time that there was a distinct lack of substantial data to support these priorities, e.g. on product demand, the feasibility of product development and deployment, or the likely impact of product use. Equally, the decision-making process here could be criticised for lack of due consideration of endemic production diseases and non-infectious health problems.

The above example of non-data-based decision-making is not unique. In fact the lack of data was becoming ever more apparent across the board in livestock development [14]. At the time of writing, BMGF now supports more than 60 programmes in the fields of animal health, genetics, enabling systems, animal nutrition and produce markets. As this investment portfolio expanded, the need for better and more reliable data became more acute for making informed management and investment decisions to select future development programmes.

A number of reports in the grey literature and elsewhere have highlighted the need for better data on the LMIC livestock sector, e.g. studies from the International Livestock Research Institute (ILRI), the International Bank for Reconstruction and Development, and the World Bank Group. Although there were a number of established international organisations engaged in collecting and disseminating livestock-related data and statistics, such as the Food and Agriculture Organization of the United Nations (FAO), the World Organisation for Animal Health (WOAH), ILRI, the World Bank Group, and the African Union–Interafrican Bureau for Animal Resources, BMGF took an alternative path. A new independent programme was established at Edinburgh University, with potential access to world-class expertise and competencies, to play a non-competitive, brokering role in bringing the livestock data community together. Its specific remit was to improve the availability, quality and reliability of livestock-related data to enable more evidence-based decision-making for livestock investments in LMICs. This became known as the Centre for Supporting Evidence-Based Interventions in Livestock (SEBI).

The new Centre developed a systematic approach, as described below, in an attempt to first understand and then address the root causes of the issues hindering data-based investment decisions.

Decisions for what? Decisions by whom? Who are the decision-makers?

Within the current scope of SEBI, decision-makers can be considered as four groups.

1) Farmers, from small-scale producers to fully commercial farming operations, clearly need information on the health and productive performance of their stock to make rational management choices and decisions.
2) Governments need information on the status and impact of issues such as livestock diseases to prioritise, implement and monitor control programmes for economic reasons. Sectors in which potential official interventions may be needed include international trade and breed improvement/artificial insemination programmes.
3) Investors need information on the status of livestock production and respective markets to make rational investment decisions. For example, animal health companies need information on numbers and types of species, prevalent diseases, the most appropriate products, regulatory systems, distribution networks, competitors and other challenges.
4) International donors need information on the highest leverage investment opportunities for the best-possible outcomes and their economic, social and

environmental contexts – for example, what impacts investment decisions may have, what consequences (intended and unintended), and how best to shape future investment strategies. This information is needed to make the most rational and effective choices about the targeting of future funding. For example, should funding be directed towards dairy genetics, foot and mouth disease control, nutrition or some combination of these? Donors may be private or public organisations and, in the latter case, may be accountable to governments and therefore also integral to Category 2, above.

Of course, none of the above categories of decision-maker is homogeneous but rather a loose grouping of organisations with similar but individual needs. For example, a small-scale producer with one or two cows will have very different data needs from a commercial dairy with 200 cows. Thus, data needs are very much dependent on the audience or 'customer' for the data.

One of SEBI's tasks was to establish, convene and lead the community of practice Livestock Data for Decisions (LD4D), whose purpose is to drive better livestock decisions through improved data and analysis. Whilst LD4D has made significant progress in bringing the livestock data community together to find solutions for the most pressing livestock data needs, one valid criticism is that the focus has been on the data collectors and analysts, i.e. the supply side of the equation, without much consideration of the decision-makers (the customers). For this reason, LD4D has extended its focus to address the demand side by aiming to identify the specific needs of key decision-makers.

Of course, the needs of the four categories of decision-makers or customer groups above are quite different. For example, farmers need information on individual animal performance, local markets, input prices, weather, etc., as well as the capability to evaluate the impact of management changes and interventions such as disease prevention and breed improvement [15]. The other three sectors need different data sets, all consisting largely of aggregated livestock data.

What is meant by data and evidence?

'Data-driven' has become a fashionable and possibly overused phrase. If information needs to be gathered for a strategic purpose, then first a specific problem, question or objective must be posed so that researchers know what data they are looking for. So, while the answer may be data driven, the question first has to be asked: what is it we are trying to solve or achieve?

Thus, when an organisation claims to employ a 'data-driven' approach, it should have first asked what its objective or end goal is. It can then make strategic decisions based on the appropriate data, analysis and interpretation.

What are data?

Data are simply an individual or set of non-contextualised numbers or observations. For data to be translated into evidence, there has to be an opinion or hypothesis or question to add context (see Table I; https://oxford-review.com/data-v-evidence/#summary) [16]. For example, a data point might be the average milk yield of cows in Tanzania. Then, one might ask, has this changed in the last 20 years? To answer, one would need contextual information or evidence. Such evidence would consist of a series of data points collected over that time period (20 years), which has been analysed and interpreted, turning it into a useful piece of knowledge that can be communicated to another party.

What is the overall objective of data processing?

In order to collect the right data, for the right metrics or indicators, a clear statement of the problem and/or objective(s) is needed. In international development, the usual method is to construct a theory of change (ToC) (Figure 1). Theory of change is an outcomes-based approach that applies critical thinking to the design, implementation and evaluation of initiatives and programmes intended to support change [17]. A ToC is a tool to help describe the problem (objective) to be addressed, the changes that are to be made and the planned

Table I

Definitions of data and evidence (modified from Dammann) [16]

Term	Description	How it is generated	Generator	Purpose
Data	Numbers, symbols, text, images, etc.	Collected from experiments, observations, field research	Data collector (person or automated)	Generation of information
Evidence	Relevant contextualised data	Comparison with standards or other reference information	Scientist, advisor to decision-/policy-makers. In the present context, could also be a farmer	Used as a basis for decision-making

activities to achieve that objective [18]. The key to this is the 'outcomes-based' approach. That is, one starts with a desired end in mind, requiring a clear understanding and definition of what is to be achieved, and then plans the necessary steps to get there (termed the 'theory of action'). It is also important to include some metrics or indicators and intermediate milestones, so that progress can be measured during the course of the programme. An example of theory of change/theory of action is shown in Figure 1.

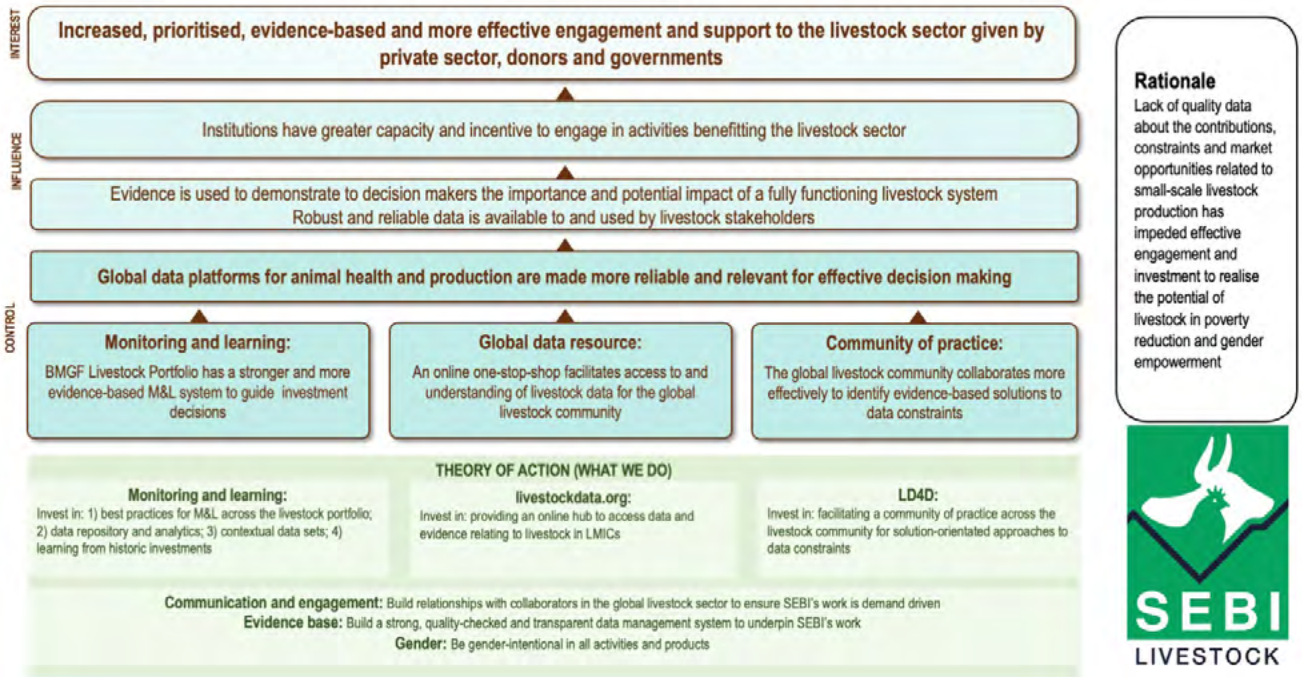
It will be clear from Figure 1 that, as the focus moves up from the bottom of the chart to the top, more context is added. It will be noted that the activities at the base of the figure are directly under the control of the acting organisation (SEBI, in this case). However, those activities are part of a much bigger context over which the acting organisation has less influence. These larger contexts are known as zones of influence, which merge into bigger zones of interest, where the acting organisation has less and less control over subsequent events. This is important when trying to determine cause and effect, because it can be extremely difficult to attribute results to any specific intervention with certainty. It is also important to differentiate the terms 'output', 'outcome' and 'impact' [19].

In the private sector, the equivalent tool would be called a business plan (BP), which essentially sets out the business's objectives and identifies the activities, processes and resources necessary to achieve them. An important component of both the ToC and BP is the use of metrics (measurements or indicators), based on real data, to monitor progress and ensure that the plan is on track and being adhered to. Thus, data are needed to measure the outputs of the programme and inform future directions. Trends in the data are also important; for example, is national milk production going up or down? Is egg production going up or down? How many more cattle are being vaccinated?

How to collect and use data to adjust course

Data gaps and flaws are well recognised in the international livestock development field [14]. While there are many sources of data on LMICs, such as FAO, WOAH, ILRI, Centre for Agriculture and Bioscience International (commonly known as CABI), World Bank Group, grey literature and peer-reviewed publications, the data are disparate and incomplete and may even be inappropriately extrapolated from the industrialised world. Data on greenhouse gas emissions in LMICs are a case in point [20, 21].

In their report 'Investing in the livestock sector: why good numbers matter', Pica-Ciamarra *et al.* [14] drew attention to



BMGF: Bill and Melinda Gates Foundation
LD4D: Livestock Data for Decisions
LMICs: low- and middle-income countries
M&L: monitoring and learning
SEBI: Centre for Supporting Evidence-Based Interventions

Figure 1
Theory of change and action for the Centre for Supporting Evidence-Based Interventions in Livestock

the deficit of good-quality data on LMIC livestock production and went on to describe the institutional necessity for the routine collection and analysis of data at an official level. However, they pointed out that such routine data are currently of poor quality, partly due to inadequate training of the extension officers who are responsible for the process. Furthermore, routine data are collated on a complete enumeration basis (in other words, a census), which makes the task extremely demanding and time consuming. For this reason, the authors advocate a statistically based sampling approach, which could make data collection more efficient and convenient. It should be noted that livestock data are not the only data lacking in many LMICs. The 50×2030 Project (https://www.50×2030.org) is a multi-lateral initiative to produce, analyse and apply data to decision-making across the whole agricultural sector, in an attempt to address these issues on an international scale.

Although there are many publications on livestock health and productivity in LMICs in the peer-reviewed and grey literature, the evidence has rarely been systematically categorised and reviewed. A key objective of SEBI is to systematically describe the available data and evidence on high-impact livestock diseases in sub-Saharan Africa. Highly detailed studies of the epidemiology of livestock disease have been carried out in limited areas [22] but these can be prohibitively expensive and are therefore unsustainable for the purpose of providing long-term routine evidence. Data collection and processing come at a cost. It is important to have reliable data and they should be fit for purpose. The law of diminishing returns can also be applied to data quality. When one starts at a low base of quality, increased investment leads to an increase in the quality and hence reliability of the data. However, as investment levels gradually increase, the improvement in quality will begin to level off, eventually leading to a point where there is little or no improvement irrespective of increased investment. Thus, the questions should be asked: how good do the data need to be? When are they good enough? When do the costs begin to outweigh the benefits?

The Centre for Supporting Evidence-Based Interventions has used a variety of methodologies to collect data on ruminant disease in several countries to assist governments in selecting priorities for intervention (https://www.livestockdata.org/data-object/farmer-surveys-and-key-informant-questionnaires-cattle-and-small-ruminant-mortality).

The Centre has also developed a protocol for systematic mapping, using Ethiopian literature on ruminant diseases over the last ten years as a pilot programme [23], and this map has now been published [24]. The aim of this study by Tsouloufi *et al.* was to collate and synthesise the published evidence on ruminant disease frequency and associated mortality in Ethiopia, by identifying knowledge gaps and clusters in the literature, to provide the basis for a decision-making tool. Search results were screened for their relevance by title, by abstract and by full text, identifying 716 articles relevant to the research question. This study suggests that, despite the high output of epidemiological publications, further understanding of a considerable number of diseases is required. Moreover, in areas where evidence is abundant, more synthesis of this information is needed to better inform decisions on disease control priorities in Ethiopia. An example of the output from the systematic map is shown in Figure 2. The output has been developed into a visual format that can be interrogated for individual species, regions, diseases, etc. and is available online at https://www.livestockdata.org.

Since carrying out such reviews is highly laborious, this process has been automated and can now be achieved with machine learning protocols [25]. The intention is to extend this process to other species, languages and countries, to add more structure to the body of data on the impact and prevalence of tropical livestock disease.

Monitoring and learning for continuous improvement

The BMGF team needs a constant flow of data in the form of 'dashboards' from funded programmes, particularly farmer-facing ones, so that their outputs can be monitored and the data aggregated across programmes, with the aim of being able to track progress, identify opportunities to improve and adjust operational directions, if necessary. These and other data coming into SEBI are cleaned and processed as described in Figure 3. Standardising the format and structure of incoming data is a work in progress.

In addition, the team depends on these data to calculate the impacts of investment and inform future investment decisions. Calculations are also carried out by SEBI on the economic returns for small-scale producers on their investments in animal health and genetics. Examples of these models are illustrated in Figures 4 and 5.

Examples of collected data and considerations to inform future directions

Investigation of mortality rates/risk

To investigate the use of a single metric – in this instance, mortality – as the measure of a country's animal health status, Wong *et al.* estimated the mortality risk in three countries: Ethiopia, Tanzania and Nigeria [26]. A further aim of this study was to determine the most prevalent diseases causing mortality with a view to identifying priorities for control. A variety of sources were used, including government institutional data, farmer recall surveys, expert opinions, living standards measurement study surveys, literature reviews, laboratory data and real-time diagnostic surveillance. All

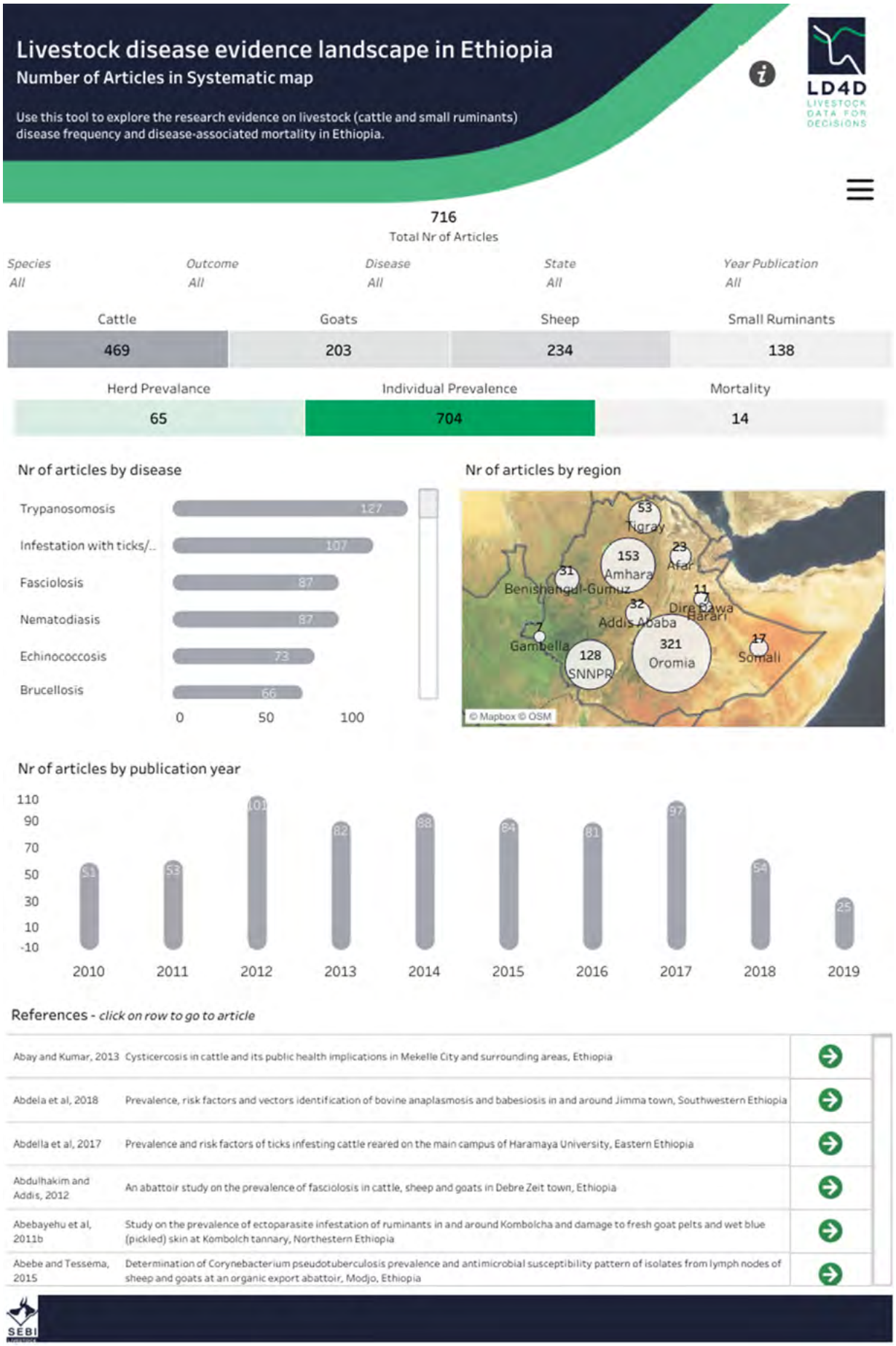


Figure 2

A static example of the output from the systematic map of literature on ruminant disease in Ethiopia [24, 25]

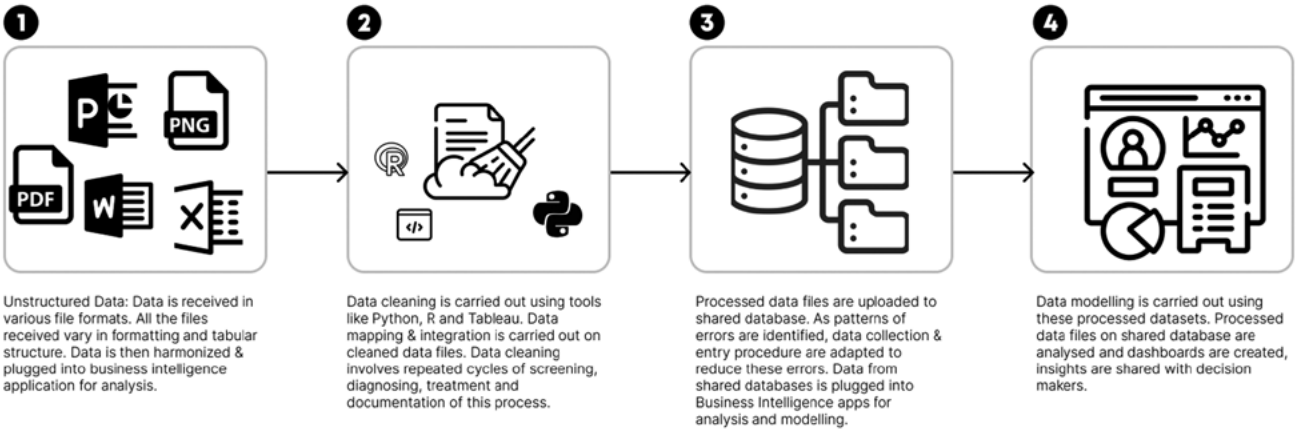


Figure 3

The data-cleaning process used by the Centre for Supporting Evidence-Based Interventions

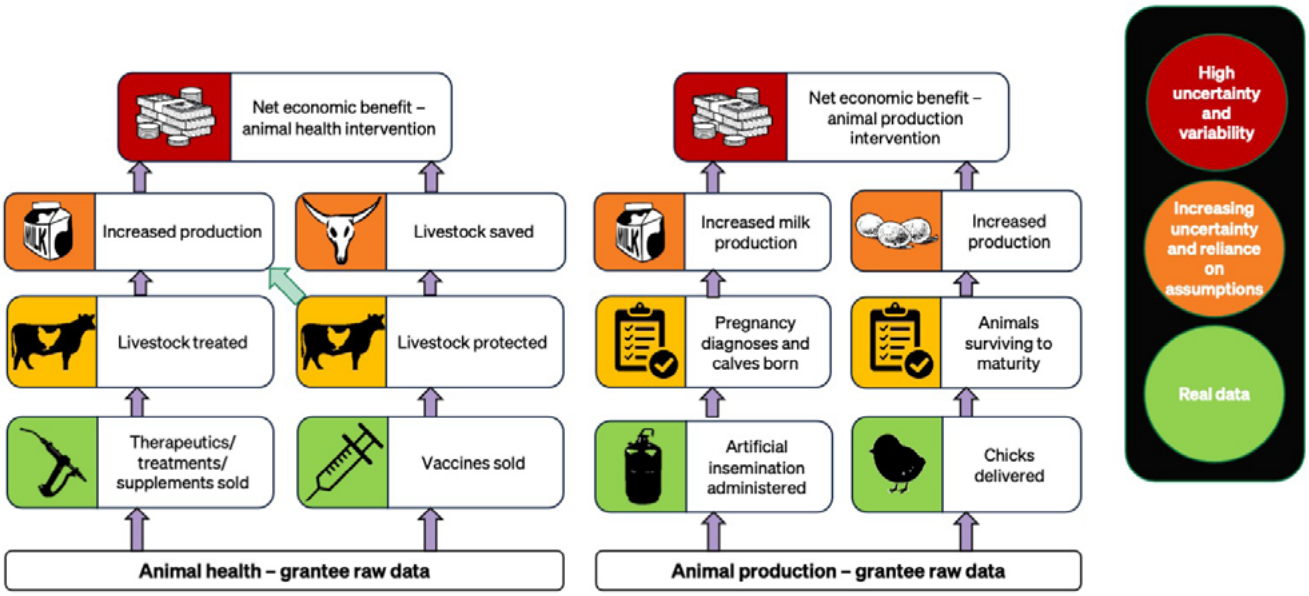
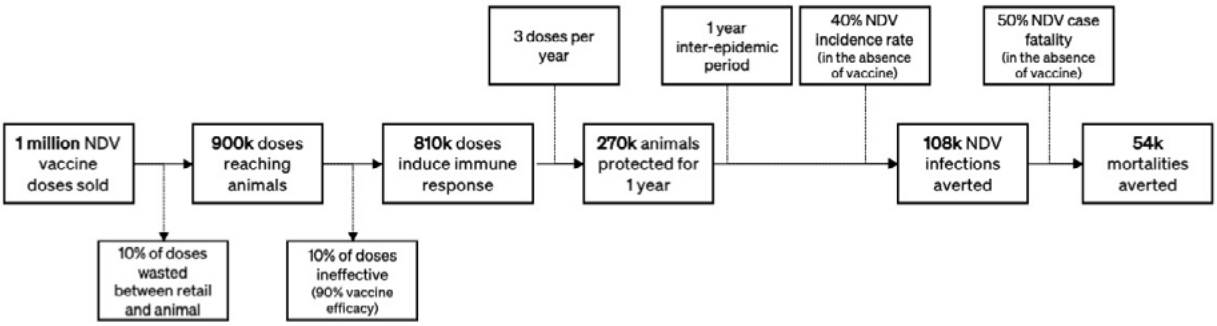


Figure 4

Simple models used to estimate the net economic benefits from using veterinary medicines or improved genetics in the livestock of small-scale producers.

Colour coding (green to red) demonstrates the increased reliance on evidence-based assumptions to estimate uncertain and variable outcomes and impact

Courtesy of Gareth Salmon, Centre for Supporting Evidence-Based Interventions



NDV: Newcastle disease vaccine

Figure 5

Simple model used to estimate the impact of Newcastle disease vaccine from the number of vaccine doses sold by a vaccine manufacturer

Courtesy of Gareth Salmon, Centre for Supporting Evidence-Based Interventions

these methods have recognised flaws [14]. It soon became apparent that the estimates within and between years varied so widely that the use of a single measure would be unachievable, particularly given the huge gaps in systematic data collection. On further analysis, since a high proportion of livestock mortalities in LMICs occur during the early weeks of life [SEBI, unpublished data], SEBI has suggested the use of a young stock mortality metric as a more reliable indicator of the overall animal health status of a country [26].

Data on national Veterinary Services

The World Organisation for Animal Health publishes regular reports on national Performance of Veterinary Services. A tool was developed to extract data from such reports so that they could be manipulated into a summary and analysable form (available at: https://www.livestockdata.org/data-object/performance-veterinary-services-pvs). This interactive visualisation can be interrogated to 'zoom in' on specific countries, years, etc. and is illustrated in Figure 6.

Data for farmers

Recordkeeping is a necessary element of good business management. With no written records, farmers have to depend on their memory when making decisions on farm practices (https://vikaspedia.in/agriculture/livestock/cattle-buffalo/importance-of-record-keeping-at-livestock-farm). The sophisticated livestock industries of the developed world have been built with the vital support of on-farm performance (phenotypic) data recording and this has usually required precursors of institutional development, such as livestock federations and related organisations, e.g. national dairy federations. The International Committee for Animal Recording is an international non-governmental organisation and network with members from 120 countries,

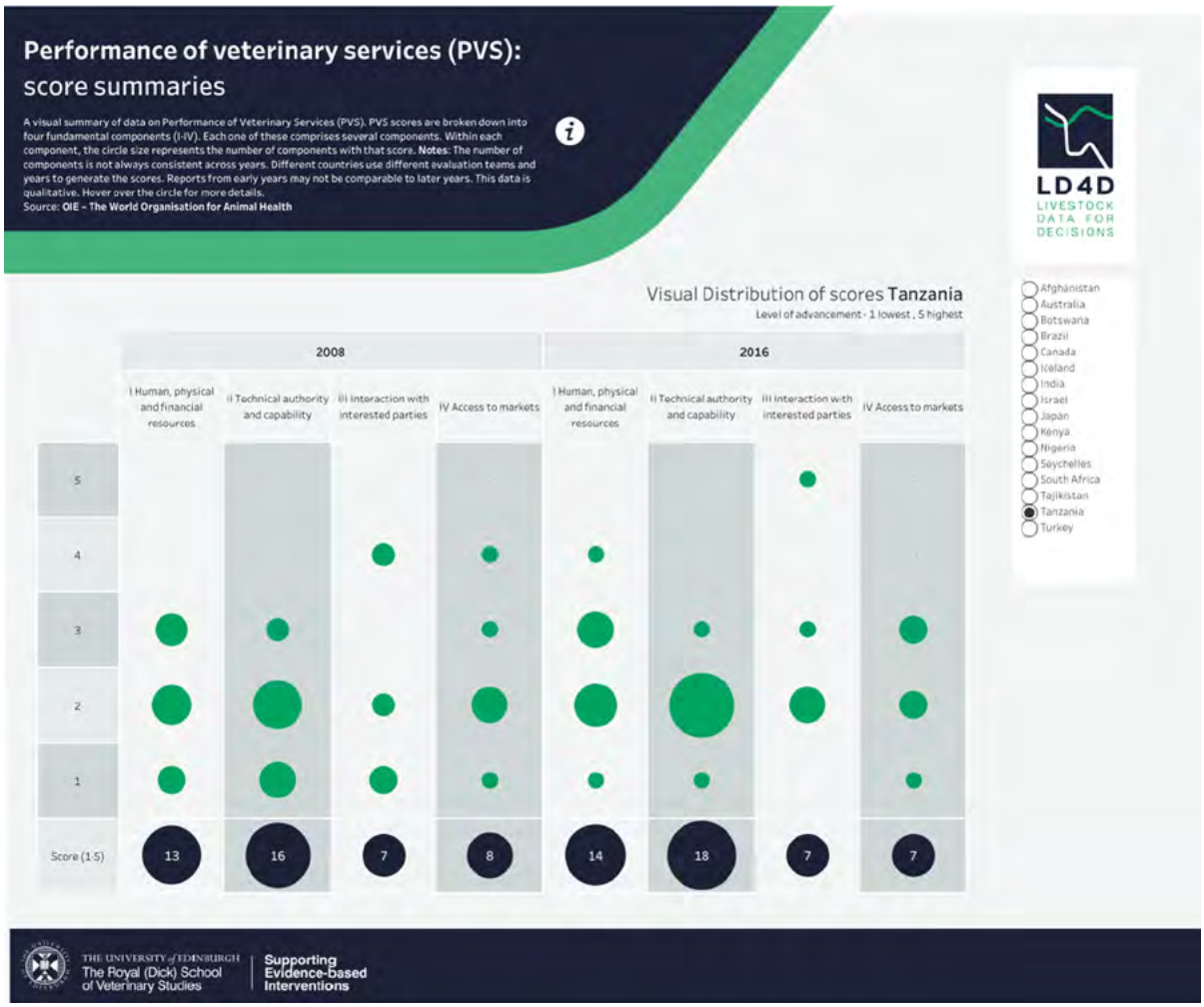


Figure 6

A sample visualised screenshot of Performance of Veterinary Services (PVS) scores for Tanzania in 2008 and 2016, from the World Organisation for Animal Health's published PVS reports

Courtesy of Louise Donnison, Centre for Supporting Evidence-Based Interventions

which aims to set standards for livestock identification, performance recording and genetic evaluation. One future aspiration would be for those bodies generating livestock data in LMICs to work towards alignment with these international standards.

While the need for good on-farm records to improve livestock performance is well recognised, clearly systems are not in place in most LMICs to fully exploit this. One of the main obstacles is the need for individual animal data and, until the advent of digital methods of data collection, this relied on 'pen and paper' or, worse, farmer recall. Digital methods of phenotypic recording are rapidly advancing in industrialised settings, including the use of activity meters, physiological measures, sound monitors, cameras, environmental monitors, oestrus detection devices, etc. The concept of precision livestock farming (PLF) was introduced during the last decade or so. Its aim is to manage individual animals by continuous real-time monitoring of their health, welfare, production/reproduction, and environmental impact [27]. A recent systematic review of PLF in the poultry sector [28] revealed that a very low proportion (4.9%) of research papers had come from LMICs. Even in high-income countries, most of these technologies are still at the prototype stage. The main limitation in LMICs is that much of the livestock production is small scale, pastoral in nature, and often in remote areas. Furthermore, the new sensing technologies and networks are not sufficiently available in convenient, low-cost forms to be widely applied [29]. The lack of a consistent and mandatory animal identification system remains a fundamental barrier. While there have been promising developments in LMICs, e.g. Stellapps (https://www.stellapps.com), which aims to provide one-stop dairy-chain digitisation via the Internet of things, and MoooFarm (https://mooo.farm), which provides digital livestock management, veterinary and other market-related information services for small-scale livestock producers in India, they are yet to be adopted at scale to achieve their promised impact. Numerous mobile phone apps have been developed for use in remote areas, e.g. iCow (https://icow.co.ke) for support in veterinary diagnostics, phenomics, market and other information services, but they remain small scale and have yet to develop sustainable business models.

Conclusions

The goal of data-driven investment and performance management in the LMIC livestock sector is a work in progress. However, it is apparent that:

- improved systems of data availability are necessary for evidence-based decision-making on livestock priorities in LMICs;
- data collection and collation must first be put into context by identifying the relevant decision-maker (farmer, investor, government, donor), then tailored to the specific question(s) being asked, so that the data compilation system is cost effective and fit for purpose;
- ideally, data collection systems should be sustainable so that they do not require continuous donor funding;
- finally, data is only one component of decision-making and all organisations have their own specific individual cultures, needs and methodologies. Other factors necessarily come into play, such as personal preferences and biases at all levels, the influence of competing strategies and policies, levels of understanding, etc. Nevertheless, if factual, data-based information is available, this can at least provide a rational basis for further discussion and evidence-based decision-making.

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Investissements et gestion des performances fondés sur les données dans le secteur de l'élevage

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Résumé

La prise de décision fondée sur des données probantes est désormais une évidence pour de nombreux secteurs, et elle devient de plus en plus importante dans la priorisation des initiatives de développement dans les pays à revenus faibles et intermédiaires. Dans le secteur du développement de l'élevage, peu de données sont disponibles concernant la santé et la production animales, or ces éléments sont nécessaires pour établir une base de données probantes. De nombreuses décisions politiques et stratégiques sont donc prises sur la base d'éléments plus subjectifs – opinions d'experts ou autres. On observe cependant une tendance en faveur d'une approche davantage ancrée dans les données pour ce type de décisions. Le *Centre for Supporting Evidence-Based Interventions in Livestock* [Centre de soutien aux interventions basées sur des données probantes dans le domaine de l'élevage] a été établi par la Fondation Bill et Melinda Gates en 2016 à Édinbourg. Il a pour objectif de rassembler et de publier les données relatives à la production et à la santé du bétail, d'animer une communauté de pratiques en vue d'harmoniser les méthodologies relatives aux données sur le bétail et d'élaborer et de superviser les indicateurs de performance utilisés pour orienter les investissements dans le secteur de l'élevage.

Mots-clés

Décisions – Décisions fondées sur des données probantes – Élevage – Gestion des performances de l'élevage – Investissement fondé sur des données – Investissements dans l'élevage – Productivité.

Inversión y gestión del rendimiento basados en datos en el sector ganadero

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Resumen

La adopción de decisiones atendiendo a datos y criterios factuales, algo que ya es un axioma en muchos sectores, ha venido cobrando creciente importancia a la hora de fijar prioridades de desarrollo en los países de renta baja o de renta media. En el sector pecuario han faltado los datos de sanidad y de producción necesarios para establecer una sólida base factual. De ahí que buena parte de los procesos decisorios de carácter estratégico y normativo hayan discurrido por el terreno más subjetivo de las opiniones, ya provengan estas de expertos o de cualquier otro ámbito. Ahora, sin embargo, se tiende cada vez más a tratar de adoptar este tipo de decisiones a partir de una base más objetiva. En 2016, la Fundación Bill y Melinda Gates estableció en Edimburgo el *Centre for Supporting Evidence-Based Interventions in Livestock* [Centro de apoyo a las intervenciones basadas en datos factuales en ganadería], cuyos objetivos se cifran en reunir y publicar datos de sanidad y producción pecuarias, encabezar una dinámica de praxis colectiva para armonizar los métodos referidos a los datos de la ganadería y elaborar y seguir de cerca los indicadores de rendimiento utilizados para orientar las inversiones en el sector pecuario.

Palabras clave

Decisiones – Decisiones fundamentadas en datos factuales – Ganado – Gestión del rendimiento ganadero – Inversiones basadas en datos – Inversiones en ganadería – Productividad.

References

[1] Arjmand P., Noshad A. & Masoodi R. (2020). – An integrated model to facilitate and increase productivity in the production of heavy livestock. *J. Altern. Vet. Med.*, **3** (6), 319–332. Available at: https://joavm.kazerun.iau.ir/article-1-30-en.html (accessed on 7 April 2022).

[2] Guyatt G., Cairns J. […] & Tugwell P. (1992). – Evidence-based medicine: a new approach to teaching the practice of medicine. *JAMA*, **268** (17), 2420–2425. https://jamanetwork.com/journals/jama/article-abstract/400956

[3] Wieten S. (2018). – Expertise in evidence-based medicine: a tale of three models. *Philos. Ethics Humanit. Med.*, **13** (1), 2. https://doi.org/10.1186/s13010-018-0055-2

[4] Vere J. & Gibson B. (2019). – Evidence-based medicine as science. *J. Eval. Clin. Pract.*, **25** (6), 997–1002. https://doi.org/10.1111/jep.13090

[5] Martini C. (2021). – What 'evidence' in evidence-based medicine? *Topoi*, **40** (2), 299–305. https://doi.org/10.1007/s11245-020-09703-4

[6] Oettinger A.P. & Zharova M.E. (2021). – What is evidence-based medicine? [in Russian]. *Russ. J. Evidence-Based Gastroenterol.*, **10** (1), 38–48. https://doi.org/10.17116/dokgastro20211001138

[7] Kastelic J.P. (2006). – Critical evaluation of scientific articles and other sources of information: an introduction to evidence-based veterinary medicine. *Theriogenology*, **66** (3), 534–542. https://doi.org/10.1016/j.theriogenology.2006.04.017

[8] Schmidt P.L. (2007). – Evidence-based veterinary medicine: evolution, revolution, or repackaging of veterinary practice? *Vet. Clin. North Am. Small Anim. Pract.*, **37** (3), 409–417. https://doi.org/10.1016/j.cvsm.2007.01.001

[9] Jarvis S. (2020). – Proposed evidence-based veterinary medicine manifesto. *Vet. Rec.*, **187** (2), e15. https://doi.org/10.1136/vr.m2931

[10] Steele M. (2020). – Evidence-based veterinary medicine: searching the evidence in practice. *Livestock*, **25** (5), 230–233. https://doi.org/10.12968/live.2020.25.5.230

[11] Baba V.V. & HakemZadeh F. (2012). – Toward a theory of evidence based decision making. *Manag. Decis.*, **50** (5), 832–867. https://doi.org/10.1108/00251741211227546

[12] Akobeng A.K. (2005). – Principles of evidence based medicine. *Arch. Dis. Child.*, **90** (8), 837–840. https://doi.org/10.1136/adc.2005.071761

[13] Perry B.D., Randolph T.F., McDermott J.J., Sones K.R. & Thornton P.K. (2002). – Investing in animal health research to alleviate poverty. International Livestock Research Institute, Nairobi, Kenya, 140 pp. Available at: https://hdl.handle.net/10568/2308 (accessed on 7 April 2022).

[14] Pica-Ciamarra U., Baker D., Morgan N., Zezza A., Azzarri C., Ly C., Nsiima L., Nouala S., Okello P. & Sserugga J. (2014). – Investing in the livestock sector: why good numbers matter, a sourcebook for decision makers on how to improve livestock data. World Bank, Washington, DC, United States of America, 132 pp. Available at: https://openknowledge.worldbank.org/handle/10986/17830 (accessed on 7 April 2022).

[15] Marshall K., Salmon G.R., Tebug S., Juga J., MacLeod M., Poole J., Baltenweck I. & Missohou A. (2020). – Net benefits of smallholder dairy cattle farms in Senegal can be significantly increased through the use of better dairy cattle breeds and improved management practices. *J. Dairy Sci.*, **103** (9), 8197–8217. https://doi.org/10.3168/JDS.2019-17334

[16] Dammann O. (2018). – Data, information, evidence, and knowledge: a proposal for health informatics and data science. *Online J. Public Health Inform.*, **10** (3), e224. https://doi.org/10.5210/OJPHI.V10I3.9631

[17] Vogel I. (2012). – Review of the use of 'Theory of Change' in international development. Department for International Development, London, United Kingdom, 86 pp. Available at: https://gsdrc.org/document-library/review-of-the-use-of-theory-of-change-in-international-development (accessed on 7 April 2022).

[18] Harries E., Hodgson L. & Noble J. (2014). – Creating your Theory of Change: NPC's practical guide. New Philanthropy Capital (NPC), London, United Kingdom, 29 pp. Available at: https://www.thinknpc.org/wp-content/uploads/2018/07/Creating-your-theory-of-change1.pdf (accessed on 7 April 2022).

[19] Belcher B. & Palenberg M. (2018). – Outcomes and impacts of development interventions: toward conceptual clarity. *Am. J. Eval.*, **39** (4), 478–495. https://doi.org/10.1177/1098214018765698

[20] Ndung'u P.W., Bebe B.O., Ondiek J.O., Butterbach-Bahl K., Merbold L. & Goopy J.P. (2020). – Improved region-specific emission factors for enteric methane emissions from cattle in smallholder mixed crop: livestock systems of Nandi County, Kenya. Corrigendum. *Anim. Prod. Sci.*, **60** (13), 1668. https://doi.org/10.1071/AN17809_CO

[21] Goopy J.P., Ndung'u P.W., Onyango A., Kirui P. & Butterbach-Bahl K. (2021). – Calculation of new enteric methane emission factors for small ruminants in western Kenya highlights the heterogeneity of smallholder production systems. *Anim. Prod. Sci.*, **61** (6), 602–612. https://doi.org/10.1071/AN19631

[22] Callaby R., Jennings A., Mwangi S.T., Mbole-Kariuki M., Van Wyk I., Kiara H., Coetzer J.A.W., Woolhouse M.E.J., Hanotte O., Toye P.G. & Bronsvoort B.M. deC. (2020). – Reflections on IDEAL: what we have learnt from a unique calf cohort study. *Prev. Vet. Med.*, **181**, 105062. https://doi.org/10.1016/j.prevetmed.2020.105062

[23] Tsoulouf T.K., Donnison L.M., Smyth K.L. & Peters A.R. (2020). – Development of a systematic mapping review protocol for the most recent evidence on ruminant infectious disease frequency and disease-associated mortality: Ethiopia as a case study. *Anim. Health Res. Rev.*, **21** (1), 96–102. https://doi.org/10.1017/s1466252319000203

[24] Tsoulouf T.K., MacVicar I.S., Donnison L.M., Smyth K.L. & Peters A.R. (2023). – Systematic map of the most recent evidence (2010–2019) on ruminant production-limiting disease prevalence and associated mortality in Ethiopia. *Front. Vet. Sci.*, **10**, 995213. https://doi.org/10.3389/fvets.2023.995213

[25] Goldfarb-Tarrant S., Robertson A., Lazic J., Tsoulouf T., Donnison L. & Smyth K. (2020). – Scaling systematic literature reviews with machine learning pipelines. *In* Proc. 1st workshop on scholarly document processing. Association for Computational Linguistics, November, online workshop, 184–195. https://doi.org/10.18653/v1/2020.sdp-1.21

[26] Wong J.T., Vance C.J. & Peters A.R. (2021). – Refining livestock mortality indicators: a systematic review. *Gates Open Res.*, **5**, 75. https://doi.org/10.12688/gatesopenres.13228.1

[27] Berckmans D. & Guarino M. (2017). – From the editors: precision livestock farming for the global livestock sector. *Anim. Front.*, **7** (1), 4–5. https://doi.org/10.2527/af.2017.0101

[28] Rowe E., Dawkins M.S. & Gebhardt-Henrich S.G. (2019). – A systematic review of precision livestock farming in the poultry sector:

is technology focussed on improving bird welfare? *Animals*, **9** (9), 614. https://doi.org/10.3390/ani9090614

[29] Mrode R., Dzivenu C.E., Marshall K., Chagunda M.G.G., Muasa B.S., Ojango J. & Okey A.M. (2020). – Phenomics and its potential impact on livestock development in low-income countries: innovative applications of emerging related digital technology. *Anim. Front.*, **10** (2), 6–11. https://doi.org/10.1093/af/vfaa002

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